Interpretive summary and conclusions

Evolution of ISUP Grade Group 1 Detection Across Prostate Cancer Screening Strategies

This document interprets the completed meta-analysis in `REPORT.md`. It is a summary of what the reported estimates support, what they do not establish, and how the main limitations affect interpretation. It is not a clinical guideline and does not replace assessment of mortality, quality of life, harms, or risk of bias.

Executive conclusion

The analysis supports a distinction between **inviting more men to screening** and **changing the diagnostic pathway after screening has begun**.

1. **Screening invitation increased low-grade cancer detection on average, but the result is heterogeneous and not robust enough to imply a predictable effect in a new setting.** Invitation versus usual care increased GG1 detection per randomized or invited participant (RR 1.614, 95% CI 1.049–2.483), yet heterogeneity was extreme (I^2 99.0%) and the prediction interval included both a reduction and a large increase (0.533–4.887).
2. **Screening invitation did not establish increased GG≥2 detection.** The pooled RR was 1.117 (95% CI 0.721–1.732), with I^2 93.6% and a prediction interval of 0.692–1.802. The evidence therefore remains compatible with lower, unchanged, or higher GG≥2 detection depending on context.
3. **The strongest and most consistent result concerns diagnostic selectivity.** Among trials in which both arms were screened, the evaluated diagnostic or biopsy strategy increased the proportion of detected cancers that were GG≥2 (RR 1.388, 95% CI 1.233–1.563; I^2 0%; prediction interval 1.179–1.634). This result remained above the null after omission of any one trial.
4. **Diagnostic strategies also improved GG≥2 yield among biopsied participants.** The pooled RR was 2.094 (95% CI 1.421–3.087), with moderate heterogeneity (I^2 44.5%) and a prediction interval above 1 (1.175–3.732). This suggests that the newer pathways direct biopsies toward participants more likely to have higher-grade disease.
5. **These findings do not show that any strategy improves survival or net clinical benefit.** The analysis concerns detection and diagnostic yield. Mortality, disease progression, quality of life, treatment burden, adverse events, and complete procedural harms were not analyzed.

The most defensible overall interpretation is therefore: **newer diagnostic pathways appear better at concentrating GG≥2 disease among cancers found and among people biopsied, whereas broad screening invitation may increase GG1 detection without clear evidence of increased GG≥2 detection.** No strategy can be declared clinically superior from these outcomes alone.

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1. What the randomized comparisons show

Screening invitation versus no screening or usual care

Outcome	Pooled result	Interpretation
GG1 per randomized/invited participant	RR 1.614 (1.049–2.483)	Average increase, but extreme inconsistency and a prediction interval crossing 1
GG1 among all detected cancers	RR 1.204 (1.003–1.444)	Small pooled increase; borderline and sensitive to individual trials
GG≥2 per randomized/invited participant	RR 1.117 (0.721–1.732)	No clear increase or decrease established
GG≥2 among all detected cancers	RR 0.823 (0.594–1.141)	No clear change established

The invitation-family results are dominated by between-trial differences: I^2 ranged from 93.6% to 99.0%. All four prediction intervals crossed the null. Consequently, the pooled values describe an average across substantially different trials rather than a stable effect expected in every screening program.

The two nominally positive GG1 findings are also fragile. Excluding CAP changed the GG1 detection estimate to RR 1.718 (95% CI 0.817–3.611), and excluding either CAP or ERSPC removed the clear increase in the GG1 share of all cancers. The mature CAP update moved the latter estimate only just above the null (RR 1.204, 95% CI 1.003–1.444). These analyses support concern that invitation can increase low-grade detection, but they do not provide a precise or broadly transportable estimate of its size.

This pattern is compatible with additional detection of indolent disease, but **overdiagnosis cannot be directly quantified from GG1 detection alone**. Overdiagnosis requires evidence that a detected cancer would not have caused symptoms or death during the person's lifetime.

Diagnostic or biopsy strategy when both arms were screened

Outcome	Pooled result	Interpretation
GG1 per randomized/invited participant	RR 0.563 (0.152–2.081)	Direction favors less GG1 detection, but highly uncertain
GG1 among all detected cancers	RR 0.567 (0.302–1.064)	Suggestive reduction, not conclusively established
GG≥2 per randomized/invited participant	RR 1.292 (0.571–2.925)	No clear population-level detection difference
GG≥2 among all detected cancers	RR 1.388 (1.233–1.563)	Consistent enrichment of GG≥2 among cancers detected
GG1 among participants biopsied	RR 0.878 (0.601–1.281)	No clear difference
GG≥2 among participants biopsied	RR 2.094 (1.421–3.087)	Higher GG≥2 yield among biopsied participants

The distinction between denominators is essential. The evidence is strongest for **who is selected for diagnosis and what grade is found**, not for an increase in the absolute number of GG≥2 cancers detected across the randomized population.

The GG≥2 share among all detected cancers is the most reliable comparative finding in the report. It had no observed heterogeneity, its prediction interval remained above 1, and every leave-one-out estimate remained above 1 with its confidence interval excluding 1. The GG≥2 biopsy-yield result points in the same direction and has a prediction interval above 1, although some leave-one-out confidence intervals crossed 1 because only three studies remained.

Conversely, the apparent reductions in GG1 outcomes are not conclusive. Their confidence and prediction intervals include no effect, and heterogeneity is substantial for GG1 detection per randomized participant. The data therefore support improved concentration of higher-grade disease more strongly than they support a definite reduction in low-grade detection.

Timing of screening initiation

PROBASE is a single timing contrast and should not be pooled conceptually with screening invitation or diagnostic-strategy trials. Its very large GG≥2 detection RR (66.885) is driven by zero inferred GG≥2 events in the delayed control arm and has an extremely wide confidence interval (4.098–1091.529). The estimate changes markedly under alternative continuity corrections.

This result indicates that more GG≥2 cancers were found at the analyzed early time point in the immediate-screening group. It does not provide a stable estimate of long-term benefit and may partly reflect earlier diagnosis rather than a lasting difference in cumulative disease detection.

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2. Descriptive strategy patterns

The separate proportion models can generate hypotheses but cannot rank strategies. They compare different studies and populations rather than randomized head-to-head assignments.

Within that limitation, three patterns are notable:

- **Biomarker → MRI pathways showed the most selective profile:** GG1 comprised 16.4% of detected cancers, GG≥2 comprised 81.9%, GG≥2 was found in 61.6% of biopsied participants, and 3.7% of screened or assessed participants underwent biopsy.
- **MRI-first pathways showed relatively high GG≥2 detection** per randomized or invited participant (4.4%) and a high GG≥2 biopsy yield (43.8%), but also the highest descriptive biopsy proportion (10.4%).
- **PSA-only estimates were highly inconsistent across studies.** I^2 was 99.3%–100% for population detection, cancer-grade composition, and biopsy use, and remained high for biopsy yield. A single pooled PSA proportion is therefore a poor prediction for a specific program.

These values should not be interpreted as proof that biomarker → MRI is superior to MRI-first, PSA → MRI, or PSA alone. Differences in trial populations, eligibility, thresholds, follow-up, adherence, and outcome ascertainment can produce the observed contrasts. Most modern-strategy estimates contain only two study-strategy units.

3. Robustness and evidential weight

Findings with the strongest support

- Diagnostic/biopsy strategies increased GG≥2 among all detected cancers: RR 1.388 (95% CI 1.233–1.563), I^2 0%, prediction interval 1.179–1.634.
- The direction and statistical conclusion were unchanged in every leave-one-out analysis.
- Diagnostic/biopsy strategies increased GG≥2 among biopsied participants: RR 2.094 (95% CI 1.421–3.087), with a prediction interval above 1.

Findings that are suggestive but uncertain

- Diagnostic strategies may reduce GG1 as a share of detected cancers, but the 95% CI narrowly includes no effect and the prediction interval is wide.
- Screening invitation may increase GG1 detection, but the result varies substantially by trial and is sensitive to exclusions.

Findings not established by this analysis

- A clear effect of screening invitation on GG≥2 detection.
- A clear effect of diagnostic strategy on GG≥2 detection across all randomized participants.
- A definitive reduction in GG1 detection from modern diagnostic pathways.
- Superiority of one named screening strategy over another.
- Reduction in prostate-cancer mortality, all-cause mortality, metastatic disease, treatment, or adverse outcomes.

4. Limitations that materially affect interpretation

1. **Few studies per model.** Comparative models contain four trials and the largest descriptive model contains five. Estimates of heterogeneity, confidence intervals, and prediction intervals are consequently imprecise.
2. **Marked clinical and statistical heterogeneity.** Invitation trials differ in era, follow-up, adherence, background PSA use, screening intensity, and diagnostic work-up. The very high I^2 values are substantive, not merely statistical noise.
3. **CAP variance is not cluster-adjusted.** CAP randomized practices, but a suitable intracluster correlation or adjusted grade-specific variance was unavailable. Its confidence interval and weight may therefore be too precise.
4. **Control-group contamination and changing practice.** PLCO had substantial opportunistic PSA screening in the control arm. Older invitation trials and newer MRI-based trials represent different diagnostic eras.

5. **Missing and inconsistent denominators.** Invitation-family biopsy-yield RRs could not be estimated. CAP biopsy counts and grade counts refer to incompatible follow-up windows, and STHLM3-MRI reports a GG≥2-specific biopsy denominator that differs from its general biopsy count.
6. **Descriptive proportions are not causal comparisons.** Strategy-level proportions are confounded by study characteristics and generally rely on only two units for newer strategies.
7. **Outcome scope is incomplete for clinical decisions.** Publication bias could not be assessed; risk-of-bias judgments, PRISMA flow, treatment, procedural harms, adverse events, and mortality outcomes are absent.

Final interpretation

The meta-analysis indicates that the principal measurable advantage of newer diagnostic pathways is **greater diagnostic concentration of GG≥2 prostate cancer**: a larger share of detected cancers and a larger share of biopsies yield higher-grade disease. The evidence is less certain that these pathways increase absolute GG≥2 detection across the randomized population or reliably reduce GG1 detection.

By contrast, broad invitation to screening appears capable of increasing GG1 detection, while a corresponding increase in GG≥2 detection is not established. Because effects varied greatly across invitation trials, the balance between additional low-grade and higher-grade detection is likely to depend strongly on the screening protocol, diagnostic pathway, adherence, background testing, and follow-up period.

The appropriate clinical inference is cautious: **diagnostic refinement appears more promising than screening expansion for improving the grade profile and yield of detected disease**, but the report cannot determine whether that improvement translates into fewer deaths, better quality of life, less overtreatment, or a favorable overall benefit–harm balance. Those conclusions require long-term clinical outcomes and a formal assessment of bias and evidence certainty.

Source: completed technical report and associated pooled-estimate and leave-one-out tables. Estimates are reproduced from the finalized analysis without recalculation.